DE–8512 **11**

DISTANCE EDUCATION

M.Sc. (Physics) DEGREE EXAMINATION, MAY 2014.

CLASSICAL AND STATISTICAL MECHANICS

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

$(5 \times 20 = 100)$

1. (a) Define generalized coordinates and obtain expression for

   (i) Generalised acceleration

   (ii) Generalised force

   (iii) Generalised potential. (8)

   (b) What is Hamilton's variation principle? (4)

   (c) Obtain Lagrange's equations of motion using D' Alemberts principle for holonomic system. (8)

2. (a) Deduce Hamilton's equation of motion for a system of particles. Hence write down the equations of motion of a particle in a central force field in space. (10)

   (b) Define Poisson bracket of two dynamical variables. (2)

   (c) Show that for any three dynamical variables u,v,w the Jacobi identity.

   $[u,[v,w]] + [v,[w,u]] + [w,[u,v]] = 0$ (8)

3. (a) Prove that the transformations $P = 1/2(p^2 + q^2)$ $Q = \tan^{-1}(q/p)$ is canonical. (6)

(b) Prove that laws of conservation of linear momentum and energy for a system of interacting particles. (6)

(c) Discuss the motion of a system as a succession of infinitesimal contact transformation. (8)

4. (a) Give an account of Hamilton Jacobi theory. (8)

(b) Prove by Hamilton – Jacobi theory that the orbit of a planet around the sun is an ellipse with sun at one of its foci. (10)

(c) Write Hamilton's equation in Poisson bracket notation. (2)

5. (a) Define Euler angles. (2)

(b) Derive Euler's equations of motion interms of Euler's angles. (8)

(c) Give the theory of symmetric top under the action of gravity. (10)

6. (a) Illustrate the problem of normal coordinates and normal frequency for a linear triatomic molecule. (10)

(b) A rigid body is rotating about an axis through the origin. Obtain relations connecting the components of total angular momentum with the components of angular velocities. (10)

7. (a) State and prove Liouville's theorem. (8)

(b) What is grand partition function? How are thermo dynamic functions related to it? (8)

(c) Show that $S = K \log W(E)$ for a micro canonical ensemble. (4)

8. (a) Discuss the statistical interpretation of entropy. (4)

(b) Illustrate the method of calculation of thermo dynamical properties of a classical ideal gas in micro canonical ensemble. (7)

(c) Define

(i) phase space

(ii) mean velocity

(iii) most probable velocity. (9)

DE–8513 **12**

DISTANCE EDUCATION

M.Sc.(Physics) DEGREE EXAMINATION, MAY 2014.

MATHEMATICAL PHYSICS

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each questions carries 20 marks.

$(5 \times 20 = 100)$

1. (a) Explain the physical meaning of orthogonal and unitary transformation. (4)

   (b) Define a Hermition matrix and obtain the condition for which the following matrix is Hermition. $\begin{pmatrix} 1 & i & 0 \\ -i & 0 & 2i \\ 0 & 2i & 2 \end{pmatrix}$. (8)

   (c) Find the eigen values and eigen vectors of the matrix $\begin{pmatrix} 2 & -2 & 3 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{pmatrix}$. (8)

2. (a) If $\Gamma n = (n-1)\Gamma(n-1)$ show that with $\Gamma 1 = 1$. $\Gamma(n+1) = n!$. (4)

   (b) Obtain the general solution of Laguerre's differential equation. (8)

   (c) Define Hermite polynomial and Derive recurrence formula

   (i) $2n\, H_{n-1}(x) = H_{(n)}'(x)$

   (ii) $2x\, H_n(x) - H_{n+1}(x) = H_n'(x)$

   (iii) $2x\, H_n(x) = 2n\, H_{n-1}(x) + H_{n+1}(x)$. (8)

3. (a) Find the solution of Bessel's differential equation $x^2\left(\frac{d^2y}{dx^2}\right)+x\left(\frac{dy}{dx}\right)+(x^2-n^2)y=0$. (10)

(b) Show that $\beta(m,n)=\frac{\Gamma m\,\Gamma n}{\Gamma m+n}$ where $m>0,\ n>0$. (5)

(c) Prove the following recurrence relation L of

(i) $xJ_n^{'}=nJ_n(x)-xJ_{n+1}(x)$

(ii) $2J_n^{'}(x)=J_{n-1}(x)-J_{n+1}(x)$. (5)

4. (a) State and prove convolution theorem. (5)

(b) Find Fourier sine and cosine transform of $f(x)$ where $f(x)=\begin{cases}1 & 0\le x<1\\ 0 & x>1\end{cases}$. (10)

(c) State and prove cauchy's integral formula $f(z_0)=\frac{1}{2\pi i}\int_C\frac{f(z)dz}{(z-z_0)}$. (5)

5. (a) Find the laplace transform of the functions $F(t)=\frac{e^{at}-1}{a}\Rightarrow\frac{e^{at}-1}{a}$. (5)

(b) State and prove cauchy residue theorem for a complex function and explain the cauchy principle value. (5)

(c) State and prove Taylor's series theorem. (10)

6. (a) Write two dimensional neat flow equation and solve it. (5)

(b) Establish the wave equation of a transverse vibration of a stretched string. (5)

(c) Apply the two dimensional wave equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial^2 y} = \frac{1}{c^2}\frac{\partial^2 u}{\partial t^2}$ to the vibration of a rectangular membrance of length a and breath b. (10)

7. (a) Show that $\frac{\partial A^i}{\partial x_i^j} + A^K \lceil i K_j$ transforms like a tenser. (7)

(b) Show that the inner product of tensor A_r^{pq} and B_r^S is tensor of rank 3. (7)

(c) Show that the characteristics of irreducible representation from on orthogonal vector system in the space of group elements. (6)

8. (a) Discuss Schmidt's orthogonalisation process for the construction of orthogonal vectors. (7)

(b) Show that (1, 0, 1) (0, 1, 1) (1, 1, 0) represent linearly independent vectors. (8)

(c) Show that $\int_{-1}^{+1} x p_n(x)\, p_{n-1}(x)\, dx = \frac{2n}{4n^2 - 1}$. (5)

DE–8514 **13**

DISTANCE EDUCATION

M.Sc. (Physics) DEGREE EXAMINATION, MAY 2014.

INTEGRATED AND DIGITAL ELECTRONICS

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Sketch the circuit of source follower and explain. At low frequency what is the maximum value of voltage gain. Give the order of magnitude of output impedance. (8)

   (b) Derive an expression for output power of an idealized class B push-pull amplifier. (8)

   (c) State and explain Miller theorem. (4)

2. (a) Draw the basic inverting amplifier with an input resistance R_1 and a feed back resistance R_f. Assuming the Op.Amp to be ideal, calculate the voltage gain of inverting amplifier. (8)

   (b) How do you solve a differential equation with constant coefficient using analog computation? (8)

   (c) Explain briefly :

      (i) CMRR

      (ii) Slew rate. (4)

3. (a) State and prove De-Morgan's law. (12)

   (b) Distinguish between counter and register. (2)

   (c) Explain with suitable diagram the functioning of simplest possible shift register. (6)

4. (a) Draw the circuit of a clocked Master-Slave J-K flip flop indicating all gates. Explain its working. How racings is avoided in this circuit. (8)

   (b) What is Karnaugh Map? Give its format. Using this map, plot a Boolean expression. (7)

   (c) Write a short note on synchronous counter. (5)

5. (a) Describe data transfer instruction of 8085 microprocessor. (8)

   (b) Elaborate the architecture of 8086 μP. (8)

   (c) What is the difference between direct and indirect addressing mode? (4)

6. (a) Explain the functioning of Photo-voltaic detector. Use suitable diagram wherever necessary. (8)

   (b) The content of a 4-bit register is initially 1101. The register is shifted 6 times to the right with serial input being 101101. What is the output of the register after each shift.? (5)

   (c) Name different processes involved in the fabrication of monolithic integrated circuit. Explain. (7)

7. (a) Explain fetch cycle and execution cycle of 8085 μP. (6)

   (b) Explain the function of program counter and stack pointer. (4)

   (c) Discuss in detail the classification of memories. (10)

8. (a) What is a solar cell? (2)

(b) Explain its types and present the IV characteristic of a solar cell with the help of neat diagram. (10)

(c) Write a short note on :

(i) Hall effect.

(ii) Resistivity measurement. (8)

DE–8515 **14**

DISTANCE EDUCATION

M.Sc. (Physics) DEGREE EXAMINATION, MAY 2014.

ELECTROMAGNETIC THEORY

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Write down Maxwell's equations in differential form and point out the physical significance of each. (7)

   (b) Obtain expression for electromagnetic wave propagation in a conducting medium. (7)

   (c) A plane travelling waves has a peak electric field $E_0 = 6V/m$. If the medium is lossless with $\in_r = 3$, $\mu_r = 1$. Find

   (i) Velocity of wave and

   (ii) Peak poynting vector. (6)

2. (a) Give an account of propagation of electromagnetic waves in anisotropic dielectric medium. (7)

   (b) A laser been has a power of 20 GW and a diameter of 2 mm. Find the peak values of E and B. (8)

   (c) Write short notes on polarisation of electromagnetic waves. (5)

3. (a) Discuss the reflection and refraction of electromagnetic waves at the interface of two non conducting media. (6)

(b) Obtain the boundary conditions satisfied by the electromagnetic field vectors $\vec{E}, \vec{D}, \vec{B}$ and $\vec{H}$ on the plane interface between two media. (7)

(c) Derive Fresnel's equations for reflection and refraction of electromagnetic waves at a plane boundary separating two dielectric media. (7)

4. (a) Deduce Brewster's law on the basis of electromagnetic theory and explain how it can be verified experimentally. (6)

(b) Determine the reflection and transmission coefficient if the incident wave is polarised with its electric power. (6)

(c) Establish the Maxwell's equations for the electromagnetic field. (8)

5. (a) What is normal and anomalous dispersion? Derive Sellmeier's equation for the refractive index of a dielectric medium. (6)

(b) Differentiate between Thomson and Rayleigh scattering. (7)

(c) Write short notes on coherence and incoherence of scattered light. (7)

6. (a) Give Lorentz theory of dispersion and show that in the case of non-polar dielectrics $\frac{n^2-1}{n^2+1}\frac{m}{\rho} = \text{constant}$ where m is the molecular weight of the dielectric, ρ its density and n the refractive index. (9)

(b) Define scattering cross section. Why does the sky appears blue? (4)

(c) Derive an expression for the guide wavelength of TE mode propagation in cylindrical waveguide. (7)

7. (a) What is a wave guide? Discuss in detail the theory of waveguides with reference to TM mode of propagation of electromagnetic waves. (9)

(b) Show that the phase velocity of electromagnetic waves becomes infinite exactly at cut off. (5)

(c) Consider the propagation of electromagnetic waves in a rectangular waveguide in TE mode. Show that there is a certain minimum frequency below with no transmission is possible. (6)

8. (a) What do you mean by a plasma? Discuss the conditions for the existence of a plasma. (5)

(b) Discuss the motion of a charged particle in

(i) a uniform electric field and

(ii) a uniform magnetic field. (7)

(c) Write down the magnetic-hydrodynamic equations and hence show that a magnetic force is equivalent to hydrostatic pressure $P_m = B^2 / 2\mu 0$ plus an additional tension along the magnetic lines of force. (8)

DE–8516 **15**

DISTANCE EDUCATION

M.Sc. (Physics) DEGREE EXAMINATION, MAY 2014.

NUMERICAL METHODS AND PROGRAMMING

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

$(5 \times 20 = 100)$

1. (a) What are the character set available in 'C' language? (5)

   (b) Explain various string functions available in 'C'. (5)

   (c) Write a 'C' program to sort the given set of 'n' numbers in ascending order. (10)

2. (a) What is a file? Explain the following file processing functions with examples. (10)

   (i) fopen

   (ii) fclose

   (iii) fscanf

   (iv) fprintf.

   (b) What is meant by array? Explain the multidimensional arrays. (10)

3. (a) Find the inverse of the given matrix by Gauss-Elimination method. (5)

$$\begin{vmatrix} 2 & 1 & 1 \\ 3 & 2 & 3 \\ 1 & 4 & 9 \end{vmatrix}$$

(b) Find the root of the equation $x^3 - 6x + 4 = 0$ using Newton-Raphson method. (10)

(c) By the Newton-Raphson method find the root of equation $e^x = 4x$ which is approximately correct to 3 places of decimal. (5)

4. (a) Write a simple 'C' program using false position method to find root of an equation. (10)

(b) Write a program in 'C' for finding the roots of a equation of the form $x^3 - 3x^2 - 5 = 0$ by Newton-Raphson method. (10)

5. (a) Solve the equation using Gauss-Elimination method. (10)

$$5x - y - 2z = 142$$
$$x - 3y - z = -30$$
$$2x - y - 3z = 5.$$

(b) The value of y is related to x by the formula $y = a + bx^n$ for the following data. Find a, b, n. (10)

x :	20	40	60	80	100
y :	5	9	14	25	36

6. (a) The following table gives the levels of prices in certain years fit a second degree parabola to the data. (10)

Year :	1875	1876	1877	1878	1879	1880
Price level :	88	87	81	78	74	79
Year :	1881	1882	1883	1884	1885	
Price level :	85	84	90	92	100	

(b) Fit a straight line of the form $y = a + bx$ following data. (5)

x :	1	2	3	4	5	6	7
y :	10.8	11.8	14.1	16.4	19.0	20.8	23.1

(c) Find the best fit of straight line. (5)

7. (a) Fit a second degree parabola to the following data taking x as the independent variable. (10)

x :	0	1	2	3	4
y :	1	1.8	1.3	2.5	6.3

Find the difference between the actual value of y and the value of y obtained from the fittest curve when $x = 2$.

(b) Fit a curve of the form $y = ae^{bx}$ to the following data. (10)

x :	0	2	4
y :	5.012	10.0	31.62

8. (a) Compute $y_{(0.3)}$ given $dy/dx + y + xy^2 = 0$, $y_{(0)} = 1$ by taking $h = 0.1$. Using Runge-Kutta method of fourth order (correct to 4 decimals). (10)

(b) Given $dy/dx = x^2 + y$, $y_{(0)} = 1$. Determine $y_{(0.02)}, y_{(0.04)}$, $y_{(0.06)}$ and $y_{(0.08)}$ using Eulers modified method. (10)

DE–8517 **21**

DISTANCE EDUCATION

M.Sc. (Physics) DEGREE EXAMINATION, MAY 2014.

SPECTROSCOPY

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks

$(5 \times 20 = 100)$

1. (a) Discuss in detail the Heitler London theory of hydrogen molecule. (10)

   (b) Explain the SP^2 hybridization (10)

2. (a) Compare and contrast the spectra of rigid and non-rigid rotators. (8)

   (b) Stress the importance of Stark effect in microwave spectroscopy. (7)

   (c) Explain the mutual exclusion principle. (5)

3. (a) How do you determine the molecular structure using IR spectroscopy? (10)

   (b) Elaborate the rotational spectra of polyatomic molecules. (10)

4. (a) Give the details regarding vibrational spectra of diatomic molecule. (10)

   (b) Write down the special features of vibrational Raman Spectra. (10)

5. Write an essay on the hyper Raman effect. (20)

6. Write a note on:

   (a) Multiphoton Spectroscopy (10)

   (b) Stimulated Raman Scattering (10)

7. (a) Define nuclear resonance. Obtain Bloch equations. (10)

   (b) Explain about Mossbauer spectroscopy and mention its applications. (10)

8. (a) Discuss about the dipole-dipole interaction.

   (b) How it is differ from spin-lattice interaction? (20)

DE–8518 **22**

DISTANCE EDUCATION

M.Sc. DEGREE EXAMINATION, MAY 2014.

Physics

QUANTUM MECHANICS

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

($5 \times 20 = 100$)

1. (a) Give an account of dual nature of matter and radiation. (4)

   (b) Establish time independent Schrodinger wave equation and discuss the significance of wave function. (8)

   (c) Explain the concept of probability current density. (8)

2. (a) Write down the postulates of Quantum Mechanics. (3)

   (b) What is a rigid rotator? Deduce expressions for eigen value and eigen function of a rigid rotator with a free axis. (10)

   (c) Write Schrodinger equation for Hydrogen atom. Separate it into radial and angular parts. (7)

3. (a) State and Prove Schwartz inequality. (8)

(b) Explain Dirac's Bra and ket rotations. (4)

(c) Applying first order perturbation theory, find out the ground state energy of the atom. (8)

4. (a) State and Prove Variation principle for obtaining approximate energies in wave mechanical problems. (8)

(b) Calculate the interaction energy of electron by variation method. (8)

(c) Write a short note on projection operator. (4)

5. (a) Write down the advantages of variation method over perturbation method. (4)

(b) Discuss WKB method for the solution of Schrodinger wave equation in a potential field and discuss its validity. (10)

(c) Write short note on

(i) Induced emission

(ii) Spontaneous emission. (6)

6. (a) Develop semi classical radiation theory and use it to determine the condition for allowed transitions. (10)

(b) What is the principle used in partial wave analysis method in scattering problem? (2)

(c) Explain briefly Rayleigh Scattering and Raman Scattering. (8)

7. (a) Define Clebsch-Gordan coefficient. (2)

(b) Obtain Clebsch–Gordan coefficient when two angular momenta $j_1 = 1/2$ and $j_2 = 1/2$ are coupled. (8)

(c) Construct Green's function for an outgoing wave in scattering problem and use it to calculate the amplitude in first order Born approximation. Obtain the condition of validity of Born approximation. (10)

8. (a) Write optical theorem. Is there any usefulness of this theorem in physics. (4)

(b) Derive the relation between Einstein's A and B coefficients for transition probabilities. (8)

(c) Prove that Pauli matrices are unitary and any two different Pauli matrices anti commute. (8)

DE–8519 **23**

DISTANCE EDUCATION

M.Sc. (Physics) DEGREE EXAMINATION, MAY 2014.

SOLID STATE PHYSICS

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.
Each question carry 20 marks.
(5 × 20 = 100)

1. (a) What are Brillouin Zones? Discuss the construction of the first three Brillouin zones for a square lattice. (6)

   (b) Describe the structure of Nacl. (6)

   (c) State reciprocal lattice. Show that fcc lattice is the reciprocal of the bcc lattice and vice versa. (8)

2. (a) Explain the thermal conductivity of metals. (8)

   (b) Write notes on heat capacity of a metal. (6)

   (c) Describe phonon momentum. (6)

3. (a) Write short note on Nano Structure. (5)

   (b) Discuss the Kronig-Penny model for the motion of an electron in a periodic potential. (10)

   (c) What is meant by the free electron gas model of metals? (5)

4. (a) What do you understand by the fermi energy? (6)

   (b) What is an intrinsic semiconductors? (6)

   (c) What are impurity states and what role do they play in determining the electrical conductivity of a doped semiconductor. (8)

5. (a) Give quantum theory of paramagnetism. (6)

(b) Discuss how one has been able to explain the paramagnetism of rare earth and iron group ions. (10)

(c) State and explain Hund rule. (4)

6. (a) What are called umklapp processes? Explain. (5)

(b) Describe and sketch the first Brillouin zones of bcc and fcc structures. (10)

(c) What is meant by the geometrical structure factor? (5)

7. (a) Explain the various thermal properties of metals. (8)

(b) What is the significance of negative effective mass? (5)

(c) Mention the importance of crystal analysis. (7)

8. (a) Explain the inelastic scattering by phonons. (10)

(b) Show that the Fermi surface for the free electron is spherical. Give its importance. (6)

(c) Calculate the critical Neel temperature of antiferromagnetism. (4)

DE–8520 **24**

DISTANCE EDUCATION

M.Sc. (Physics) DEGREE EXAMINATION, MAY 2014.

NUCLEAR AND PARTICLE PHYSICS

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

$(5 \times 20 = 100)$

1. (a) What do you understand by internal conversion? How does it take place. (5)

   (b) What is β decay? Give the fermi theory of β-decay and obtain the expression for emitted β-ray spectrum (15)

2. (a) What are magic numbers? How does the shell model explain the existance of magic number? (6)

   (b) What are Schmidt lines? Explain. (6)

   (c) Explain salient features of collective model of nucleus. (8)

3. (a) What are the general characteristics of nuclear forces? (5)

   (b) Give brief account of the Meson theory of nuclear forces. (10)

   (c) Write a note on the charge independence of nuclear forces. (5)

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4. (a) Derive the Breit-Wigner dispersion formula for resonance scattering. (10)

   (b) Explain the compound nuclear model and nuclear reactions. (10)

5. (a) Briefly explain the classification of the elementary particles. (10)

   (b) Discuss SU(3) classification scheme of elementary particles. Show how it has been useful in tracing the missing particles. (10)

6. (a) Write a note on nuclear isomerism. (5)

   (b) Indentify the difference between nuclear shell model and liquid drop model. (5)

   (c) Give an account of the phenomena of nuclear fission and discuss the Bohr-wheeler theory to explain its mechanism. (10)

7. (a) Explain LPT theorem. (5)

   (b) What are thermonuclear reactiors. (5)

   (c) Give the theory of n-p scattering at low incident energies. (10)

8. (a) Explain why nuclei cannot have permanent electric dipolemoment (4)

   (b) Derive Geigner-Nuttal law from the theory of α – decay. (6)

   (c) Write a note on the liquid drop model of the nucleus. How does this model account for the phenomena of nuclear fission. (10)

DE–8521 **25**

DISTANCE EDUCATION

M.Sc. (Physics) DEGREE EXAMINATION, MAY 2014.

MATERIALS SCIENCE

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Define 'creep'. Explain its phases and mechanism. (12)

   (b) Describe the visco elastic behaviour of a material. (8)

2. (a) Explain population inversion and show how it can be achieved. (8)

   (b) With a necessary diagram explain the construction and working of CO_2 laser. (12)

3. (a) Write a note on compound semiconductor laser. (8)

   (b) Explain the working of an electro-optic modulator using Kerr effect. (12)

4. (a) Discuss in detail the sputtering methods and mention their advantages. (12)

   (b) Describe the principle, theory and working of diffusion pump. (8)

5. (a) Explain Quartz crystal method of measuring the thickness of thin films in detail. (12)

   (b) What are copper ion conductors and explain its applications? (8)

6. (a) Explain the principle and working of Pirani gauge. (12)

(b) Write a detailed note on fuel cells. (8)

7. (a) Write a note on thermal evaporation. (10)

(b) Write a note on Pockel's effect. (10)

8. (a) Write an essay on LB films and its applications. (12)

(b) Write a note on solid state battery. (8)